Year 13 Maths - Normal distribution

Similar questions - Percentiles, totals and the shape of the curve

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The masses, in grams, of apples of a certain kind are modelled by $\mathrm{N}(80, \sigma^2)$. It is given that 40% of the apples have masses between 75 g and 85 g.

(a) Find σ to 3 significant figures. **[4]**

(b) Find the probability that a randomly chosen apple has mass greater than 90 g. **[2]**

2 The masses, in grams, of oranges are modelled by $\mathrm{N}(40, 25)$. Oranges are packed in bags of 8.

(a) Find the probability that a bag of 8 randomly chosen oranges has a total mass of less than 300 g. **[4]**

(b) State one assumption needed for your answer to part (a). **[1]**

(c) The graph of a normal distribution has points of inflection at $x = -1$ and $x = 5$. Write down the mean and the standard deviation, explaining your method. **[2]**

Question 1

(a) By symmetry $\mathrm{P}(A < 85) = 0.5 + 0.2 = 0.7$, so $\dfrac{5}{\sigma} = 0.5244$ and $\sigma = 9.53$

(b) $\mathrm{P}(A > 90) = \mathrm{P}\left(Z > \dfrac{10}{9.53}\right) = 0.1471$

Question 2

(a) Total $T \sim \mathrm{N}(8 \times 40,\, 8 \times 25) = \mathrm{N}(320, 200)$. $\mathrm{P}(T < 300) = \mathrm{P}\left(Z < \dfrac{-20}{\sqrt{200}}\right) = \mathrm{P}(Z < -1.414) = 0.0786$

(b) The masses of the oranges in a bag are independent

(c) The mean is midway between the points of inflection: $\mu = 2$. The inflection points are at $\mu \pm \sigma$, so $\sigma = 3$